

No.	Time	Source	Destination	Protocol	Length	Info
2439	23.378742	10.100.223.122	23.2.77.119	HTTP	308	GET / HTTP/1.1
2447	23.412233	23.2.77.119	10.100.223.122	HTTP	398	HTTP/1.1 200 OK (application/pkix-cert)
2461	23.515643	10.100.223.122	23.2.77.119	HTTP	308	GET / HTTP/1.1
2467	23.555987	23.2.77.119	10.100.223.122	HTTP	299	HTTP/1.1 200 OK (application/pkix-cert)
2477	23.565105	10.100.223.122	146.190.62.39	HTTP	518	GET / HTTP/1.1
2559	23.808573	146.190.62.39	10.100.223.122	HTTP	1390	HTTP/1.1 200 OK (text/html)
2545	23.842757	10.100.223.122	146.190.62.39	HTTP	370	GET /js/init.min.js HTTP/1.1
2673	24.077086	146.190.62.39	10.100.223.122	HTTP	480	HTTP/1.1 200 OK (application/javascript)
2675	24.085330	10.100.223.122	146.190.62.39	HTTP	388	GET /css/style.min.css HTTP/1.1
2676	24.085648	10.100.223.122	146.190.62.39	HTTP	393	GET /css/style-wide.min.css HTTP/1.1
2685	24.327664	146.190.62.39	10.100.223.122	HTTP	972	HTTP/1.1 200 OK (text/css)
2699	24.328244	10.100.223.122	146.190.62.39	HTTP	395	GET /css/style-normal.min.css HTTP/1.1
2784	24.570054	146.190.62.39	10.100.223.122	HTTP	1474	[TCP Retransmission] HTTP/1.1 200 OK (text/css)
2786	24.570054	146.190.62.39	10.100.223.122	HTTP	1182	HTTP/1.1 200 OK (text/css)
2798	24.969485	10.100.223.122	146.190.62.39	HTTP	455	GET /css/images/banner.svg HTTP/1.1
2799	24.969583	10.100.223.122	146.190.62.39	HTTP	470	GET /css/images/header-major-on-light.svg HTTP/1.1
2800	24.970149	10.100.223.122	146.190.62.39	HTTP	469	GET /css/images/header-major-on-dark.svg HTTP/1.1
2993	25.193123	146.190.62.39	10.100.223.122	HTTP/X.	1337	HTTP/1.1 200 OK
2995	25.194365	10.100.223.122	146.190.62.39	HTTP	428	GET /favicon.ico HTTP/1.1
2997	25.200100	146.190.62.39	10.100.223.122	HTTP/X.	1331	HTTP/1.1 200 OK
2998	25.200100	146.190.62.39	10.100.223.122	HTTP/X.	1381	HTTP/1.1 200 OK
3024	25.491799	146.190.62.39	10.100.223.122	HTTP	907	HTTP/1.1 200 OK (image/x-icon)
3073	27.571460	10.100.223.122	23.46.230.140	HTTP	165	GET /connecttest.txt HTTP/1.1
3078	27.624781	23.46.230.140	10.100.223.122	HTTP	241	HTTP/1.1 200 OK (text/plain)

Request Packet:

Packet 1: Occurs in the Physical Layer. We can see the raw frame here.

▼	Frame 2461: Packet, 308 bytes on wire (2464 bits), 308 bytes captured (2464 bits) on
	Section number: 1
▶	Interface id: 0 (\Device\NPF_{BC57C95B-F2BD-4BB8-9681-48FF441269C4})
	Encapsulation type: Ethernet (1)
	Arrival Time: Nov 8, 2025 09:58:43.854531000 Bangladesh Standard Time
	UTC Arrival Time: Nov 8, 2025 03:58:43.854531000 UTC
	Epoch Arrival Time: 1762574323.854531000
	[Time shift for this packet: 0.000000000 seconds]
	[Time delta from previous captured frame: 636.000 microseconds]
	[Time delta from previous displayed frame: 103.410000 milliseconds]
	[Time since reference or first frame: 23.515643000 seconds]
	Frame Number: 2461
	Frame Length: 308 bytes (2464 bits)
	Capture Length: 308 bytes (2464 bits)
	[Frame is marked: False]
	[Frame is ignored: False]
	[Protocols in frame: eth:ethertype:ip:tcp:http]
	Character encoding: ASCII (0)
	[Coloring Rule Name: HTTP]
	[Coloring Rule String: http tcp.port == 80 http2]

Packet 2: Occurs in the DataLink Layer. We can see the addition of a MAC address here. SMAC(cc:15:31:d5:3c:de), DMAC(c0:f6:ec:90:a9:0b)

```
▼ Ethernet II, Src: Intel_d5:3c:de (cc:15:31:d5:3c:de), Dst: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
  ▼ Destination: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0. .... = IG bit: Individual address (unicast)
  ▶ Source: Intel_d5:3c:de (cc:15:31:d5:3c:de)
    Type: IPv4 (0x0800)
    [Stream index: 0]
```

Packet 3: This takes place at the Network Layer, where the packet includes the source and destination IP addresses, along with additional information such as packet length, checksum, and other related details.

SIP : 10.100.223.122, DIP: 23.2.77.119

```
Internet Protocol Version 4, Src: 10.100.223.122, Dst: 23.2.77.119
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 294
    Identification: 0x8a63 (35427)
  ▶ 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.100.223.122
    Destination Address: 23.2.77.119
    [Stream index: 35]
```

Packet 4: This process takes place at the Transport Layer, where the port addresses are added. The SPA is 59063 and the destination port is 80. Additionally, sequence numbers, control bits, and the window size are also included at this stage.

```

Transmission Control Protocol, Src Port: 59063, Dst Port: 80, Seq: 1, Ack: 1, Len: 254
  Source Port: 59063
  Destination Port: 80
  [Stream index: 26]
  [Stream Packet Number: 4]
  ▶ [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 254]
  Sequence Number: 1      (relative sequence number)
  Sequence Number (raw): 3191272011
  [Next Sequence Number: 255      (relative sequence number)]
  Acknowledgment Number: 1      (relative ack number)
  Acknowledgment number (raw): 1211711610
  0101 .... = Header Length: 20 bytes (5)
  ▶ Flags: 0x018 (PSH, ACK)
  Window: 255
  [Calculated window size: 65280]
  [Window size scaling factor: 256]
  Checksum: 0x4f70 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  ▶ [Timestamps]
  ▶ [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (254 bytes)

```

Packet 5: This occurs at the Application Layer, where the GET method is used in the request line. It includes the host or domain name of the server, along with details such as accepted language, encoding, and the user agent that identifies the browser. The connection indicates whether the session should remain active or be closed after the exchange.

```

Hypertext Transfer Protocol
  ▶ GET / HTTP/1.1\r\n
  Host: r3.i.lencr.org\r\n
  Connection: keep-alive\r\n
  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, li
  Accept-Encoding: gzip, deflate\r\n
  Accept-Language: en-US,en;q=0.9\r\n
  \r\n
  [Response in frame: 2467]
  [Full request URI: http://r3.i.lencr.org/]

```

Response Packet:

Packet 1: Happens in the Physical Layer

```
▼ Frame 2467: Packet, 299 bytes on wire (2392 bits), 299 bytes captured (2392 bits) on i
  Section number: 1
  ▶ Interface id: 0 (\Device\NPF_{BC57C95B-F2BD-4BB8-9681-48FF441269C4})
  Encapsulation type: Ethernet (1)
  Arrival Time: Nov  8, 2025 09:58:43.894875000 Bangladesh Standard Time
  UTC Arrival Time: Nov  8, 2025 03:58:43.894875000 UTC
  Epoch Arrival Time: 1762574323.894875000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 0.000000000 seconds]
  [Time delta from previous displayed frame: 40.344000 milliseconds]
  [Time since reference or first frame: 23.555987000 seconds]
  Frame Number: 2467
  Frame Length: 299 bytes (2392 bits)
  Capture Length: 299 bytes (2392 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:http:pkix-cert:x509sat:x509sat:x509sat:x5
  Character encoding: ASCII (0)
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http || tcp.port == 80 || http2]
```

Packet 2: There are source and destination MAC address so it happens in the Datalink Layer.
SMAC(c0:f6:ec:90:a9:0b), DMAC(cc:15:31:d5:3c:de)

```
▼ Ethernet II, Src: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b), Dst: Intel_d5:3c:de (cc:15:31:d5:3c:de)
  ▼ Destination: Intel_d5:3c:de (cc:15:31:d5:3c:de)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ...0 .... = IG bit: Individual address (unicast)
  ▼ Source: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ...0 .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
```

Packet 3: Happens in the Network Layer. It carries the source and destination IP addresses.
SIP : 23.2.77.119, DIP: 10.100.223.122

```
▼ Internet Protocol Version 4, Src: 23.2.77.119, Dst: 10.100.223.122
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 285
    Identification: 0x13cf (5071)
  ▶ 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 51
    Protocol: TCP (6)
    Header Checksum: 0xe4b4 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 23.2.77.119
    Destination Address: 10.100.223.122
    [Stream index: 35]
```

Packet 4: We can see port addresses but they are flipped from the request packet. This happens in the Transport layer.

The SPA is 80 and the destination port is 59063

```
▼ Transmission Control Protocol, Src Port: 80, Dst Port: 59063, Seq: 1421, Ack: 255, L
  Source Port: 80
  Destination Port: 59063
  [Stream index: 26]
  [Stream Packet Number: 7]
  ▶ [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 245]
  Sequence Number: 1421      (relative sequence number)
  Sequence Number (raw): 1211713030
  [Next Sequence Number: 1666      (relative sequence number)]
  Acknowledgment Number: 255      (relative ack number)
  Acknowledgment number (raw): 3191272265
  0101 .... = Header Length: 20 bytes (5)
  ▶ Flags: 0x018 (PSH, ACK)
  Window: 501
  [Calculated window size: 64128]
  [Window size scaling factor: 128]
  Checksum: 0xd054 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  ▶ [Timestamps]
  ▶ [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (245 bytes)
  TCP segment data (245 bytes)
```

Packet 5: We have a status line with version, status code, status message. The other information is related to the content: its type, length, encoding and caching rules.

```
▼ Hypertext Transfer Protocol
  ▶ HTTP/1.1 200 OK\r\n
  Server: nginx\r\n
  Content-Type: application/pkix-cert\r\n
  Last-Modified: Mon, 28 Apr 2025 19:24:15 GMT\r\n
  ETag: "680fd5df-51a"\r\n
  Content-Disposition: attachment; filename="R3.der"\r\n
  Accept-Ranges: bytes\r\n
  Vary: Accept-Encoding\r\n
  Content-Encoding: gzip\r\n
  ▶ Content-Length: 1253\r\n
  Cache-Control: max-age=3600\r\n
  Expires: Sat, 08 Nov 2025 04:58:44 GMT\r\n
  Date: Sat, 08 Nov 2025 03:58:44 GMT\r\n
  Connection: keep-alive\r\n
  \r\n
  [Request in frame: 2461]
  [Time since request: 40.344000 milliseconds]
  [Request URI: /]
  [Full request URI: http://r3.i.lencr.org/]
  Content-encoded entity body (gzip): 1253 bytes -> 1306 bytes
  File Data: 1306 bytes
```